# Measurement Based Quantum Computation

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March 10, 2026

# Quantum Teleportation

Input state: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$. Initially, we have a 3-qubit state on $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$

$$|\Psi\rangle = |\psi\rangle \otimes \frac{1}{\sqrt{2}}\left(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle\right)$$

Alice / Bob

# Quantum Teleportation II

- Alice performs a unitary transformation $U = (H \otimes \mathbb{I})CNOT$. The state becomes

$$|\Psi'\rangle = |00\rangle|\psi\rangle + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)$$

- We perform the measurement on Alice's two qubits. Based on the outcome
  - $|00\rangle$: $\mathbb{I}$
  - $|01\rangle$: $X$
  - $|10\rangle$: $Z$
  - $|11\rangle$: $ZX$

# Outline

Preliminary: Quantum Teleportation

Cluster State

Construction of Quantum Gates

Quantum Circuit

# Cluster State: Definition

A state on a cluster $\mathcal{C}$ is a cluster state, denoted by $|\phi_{\{\kappa\}}\rangle_{\mathcal{C}}$ follows

$$K^{(a)} = \sigma_x^{(a)} \bigotimes_{b \in nbgh(a)} \sigma_z^{(b)}$$

$$K^{(a)} |\phi_{\{\kappa\}}\rangle_{\mathcal{C}} = (-1)^{\kappa_a} |\phi_{\{\kappa\}}\rangle_{\mathcal{C}} \quad \forall a \in \mathcal{C}$$

- All states follow this definition that differs by at least one eigenvalue from each other form an orthogonal basis on this $2^{|\mathcal{C}|}$-qubit Hilbert space $\{|\phi_{\{\kappa\}}\rangle_{\mathcal{C}} \,|\{\kappa\} \in \{0,1\}^{|\mathcal{C}|}\}$.

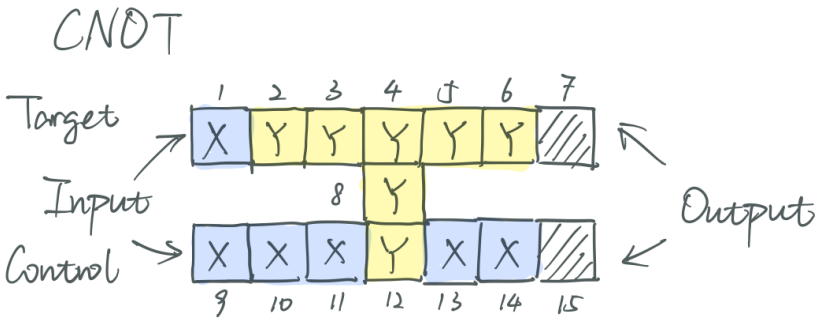
# Cluster State: Preparation

- Prepare a product state $|+\rangle_{\mathcal{C}} = \bigotimes_{a\in\mathcal{C}} |+\rangle_a$
- Apply a unitary transformation $S^{(\mathcal{C})} = \prod_{a,b\in\mathcal{C}|b-a\in\gamma_d} S^{ab}$ where $S^{ab} = \frac{1}{2}\left(1 + \sigma_z^{(a)} + \sigma_z^{(b)} - \sigma_z^{(a)} \otimes \sigma_z^{(b)}\right)$
  - The state $|1\rangle_a \otimes |1\rangle_b$ obtain a phase of $\pi$ and the rest of the states remain the same.

# Cluster State: Examples

$$
\begin{aligned}
|\phi_{\{\kappa\}}\rangle_{\mathcal{C}2} &= \frac{1}{\sqrt{2}}\left(|0\rangle_1 |+\rangle_2 + |1\rangle_1 |-\rangle_2\right) \\
|\phi_{\{\kappa\}}\rangle_{\mathcal{C}3} &= \frac{1}{\sqrt{2}}\left(|+\rangle_1 |0\rangle_2 |+\rangle_3 + |-\rangle_1 |1\rangle_2 |-\rangle_3\right) \\
|\phi_{\{\kappa\}}\rangle_{\mathcal{C}4} &= \frac{1}{2}(|+\rangle_1 |0\rangle_2 |+\rangle_3 |0\rangle_4 + |+\rangle_1 |0\rangle_2 |-\rangle_3 |1\rangle_4 \\
&\quad + |-\rangle_1 |1\rangle_2 |-\rangle_3 |0\rangle_4 + |-\rangle_1 |1\rangle_2 |+\rangle_3 |1\rangle_4)
\end{aligned}
$$

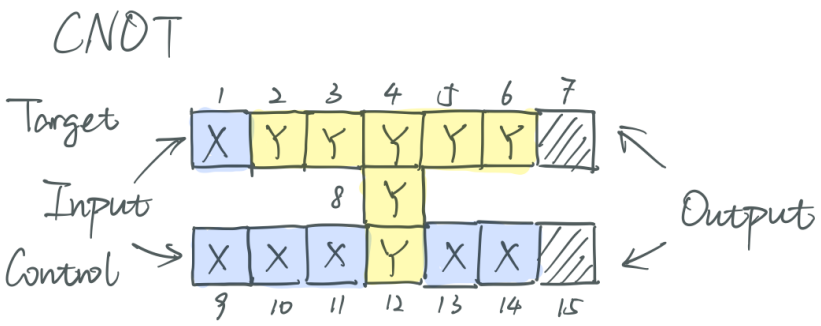
# Construction of Quantum Gates: CNOT




CNOT gate can be realized on a 15 qubit cluster state

- Prepare the initial state $|\psi_{in}\rangle_{1,9} \left( \bigotimes_{i \in \mathcal{C}_{15 \setminus \{1,9\}}} |+\rangle_i \right)$
- Entangle the 15 qubit cluster with the unitary transformation $S^{(\mathcal{C}_{15})}$
- One qubit measurement is performed on every qubit in the cluster except for the output qubit(7, 15).
  - $\sigma_x$ measurement: qubit 1, 9, 10, 11, 13, 14
  - $\sigma_y$ measurement: qubit 2, 3, 4, 5, 6, 8, 12

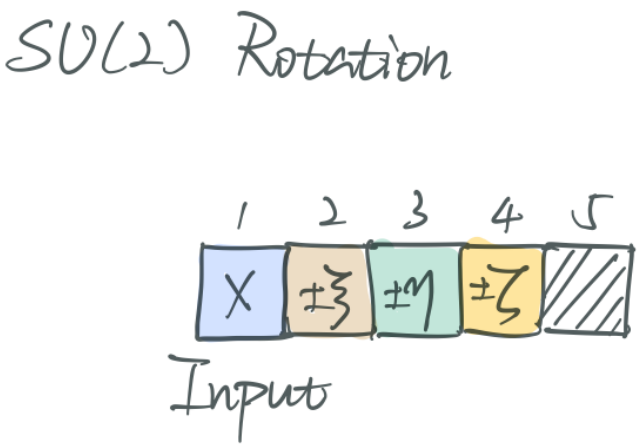
# CNOT II




- The measured qubits would be destroyed and resulting gate is $U'_{CNOT} = U_{\Sigma,CNOT}\, CNOT(c,t)$
- $U_{\Sigma,CNOT} = \sigma_x^{(c)^{\gamma_x^{(c)}}} \sigma_x^{(t)^{\gamma_x^{(t)}}} \sigma_z^{(c)^{\gamma_z^{(c)}}} \sigma_z^{(t)^{\gamma_z^{(t)}}}$ with

$$\gamma_x^{(c)} = s_2 + s_3 + s_5 + s_6$$
$$\gamma_x^{(t)} = s_2 + s_3 + s_8 + s_{10} + s_{12} + s_{14}$$
$$\gamma_z^{(c)} = s_1 + s_3 + s_4 + s_5 + s_8 + s_9 + s_{11} + 1$$
$$\gamma_z^{(t)} = s_9 + s_{11} + s_{13}$$

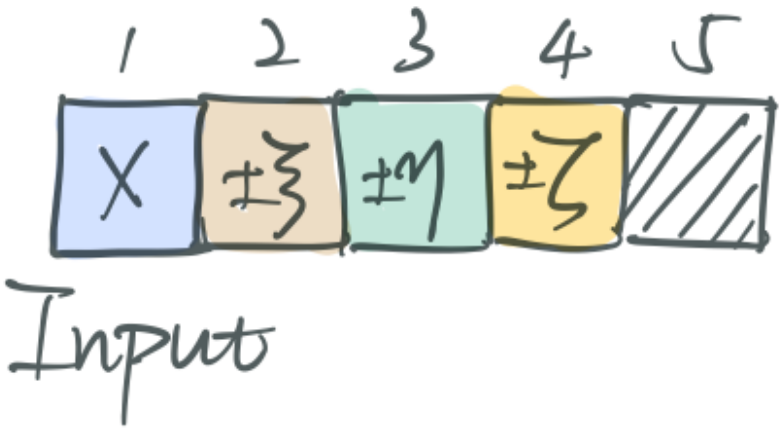
# Construction of Quantum Gates: One-Qubit Rotation




A general $SU(2)$ rotation $U_{Rot}[\xi, \eta, \zeta] = U_X[\zeta] U_Z[\eta] U_X[\xi]$ can be realized on 5-qubit cluster

- Prepare the state $|\Psi\rangle_{\mathcal{C}_5} = |\psi_{in}\rangle_1 \left( \bigotimes_{i=2}^{5} |+\rangle_i \right)$
- Create the cluster state by performing the unitary operation $S^{(\mathcal{C}_5)}$

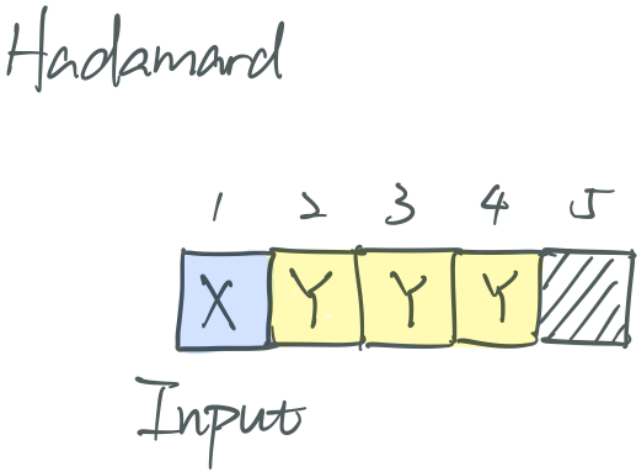
# Construction of Quantum Gates: One-Qubit Rotation

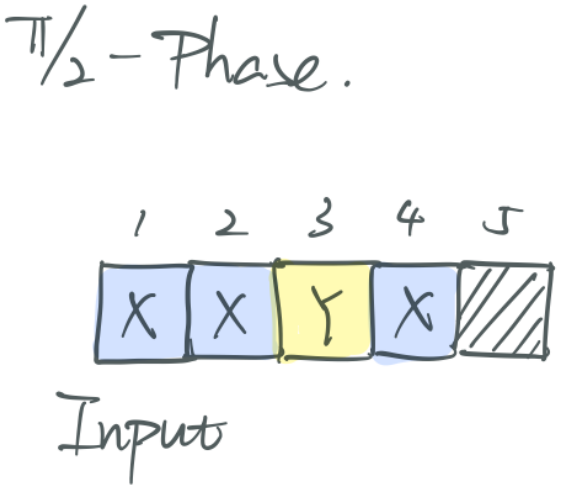



- Measure the first 4 qubits in the following order. Each measurement correspond to a step of Euler rotation
  - Measure qubit 1 in $\mathcal{B}_1(0)$.
  - Measure qubit 2 in $\mathcal{B}_2(-\xi(-1)^{s_1})$.
  - Measure qubit 3 in $\mathcal{B}_3(-\eta(-1)^{s_2})$.
  - Measure qubit 4 in $\mathcal{B}_4(-\zeta(-1)^{s_2+s_3})$.

  With $\mathcal{B}_j(\varphi_j) = \left\{ \frac{|0\rangle_i + e^{i\varphi}|1\rangle_j}{\sqrt{2}}, \frac{|0\rangle_i - e^{i\varphi}|1\rangle_j}{\sqrt{2}} \right\}$
- Then, we have performed a rotation $U'_{Rot} = U_{\Sigma,Rot} U_{Rot}[\eta, \xi, \zeta]$, with the byproduct operator $U_{\Sigma,Rot} = \sigma_x^{s_2+s_4} \sigma_z^{s_1+s_3}$

# Construction of Quantum Gates: Hadamard- and $\pi/2$-Phase Gate



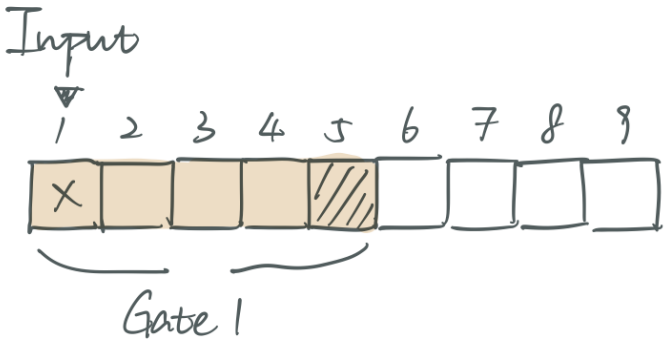



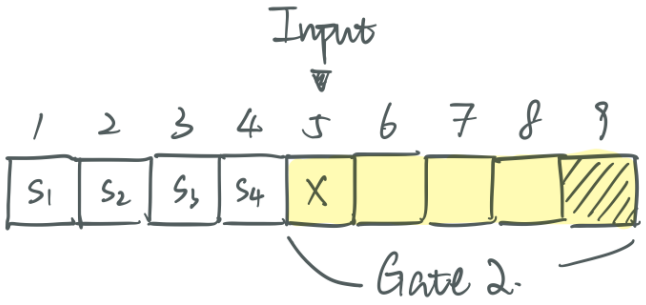

Hadamard and $\pi/2$-phase gate are two special cases of $SU(2)$ rotation. Hence, they can also be realized through 5-qubit cluster.

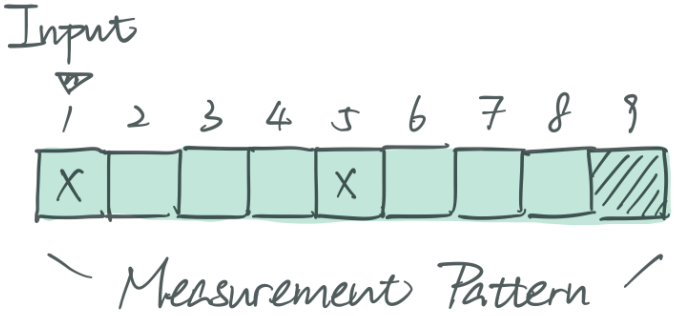
- Prepare the 5-qubit cluster state
- Measurement of the first 4 qubit can be performed simultaneously
  - Hadamard Gate: $\sigma_x^{(1)}$, $\sigma_y^{(2)}$, $\sigma_y^{(3)}$, $\sigma_y^{(4)}$, $U_{\Sigma,H} = \sigma_x^{s_1+s_3+s_4}\sigma_z^{s_2+s_3}$
  - $\pi/2$-phase Gate: $\sigma_x^{(1)}$, $\sigma_x^{(2)}$, $\sigma_y^{(3)}$, $\sigma_x^{(4)}$, $U_{\Sigma,H} = \sigma_x^{s_2+s_4}\sigma_z^{s_1+s_2+s_3+1}$.

# Quantum Circuit: Logical Network

How to combine quantum gates to quantum logical network?




**Instead of quantum gates, we have measurement patterns!**

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